Lab 5.1 – ID and Filters

At the end of this lab exercise, Participants will be able to:

1. Identify a 2D barcode on the demo plate
2. Read the alphanumeric string on the camera body
3. Use the Image Filter to enhance text

The Participant will utilize the following In-Sight Functions to successfully complete this exercise:

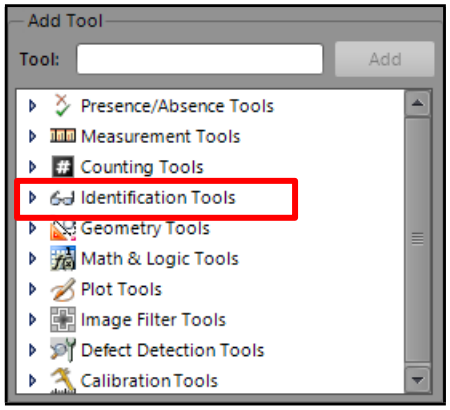
4. Easy Builder User Interface Application Step 2
   Inspect Part
   1. Identification Tools
   2. Image Filter Tools

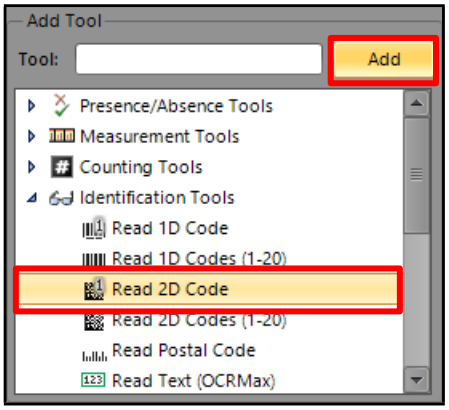
Follow the steps below to complete the lab exercise:

1. Open the **MyCount** job from the previous lab exercise.
2. Save the job as **My2D** in the folder created in Lab #2.
   Continue the **Inspect Part** 

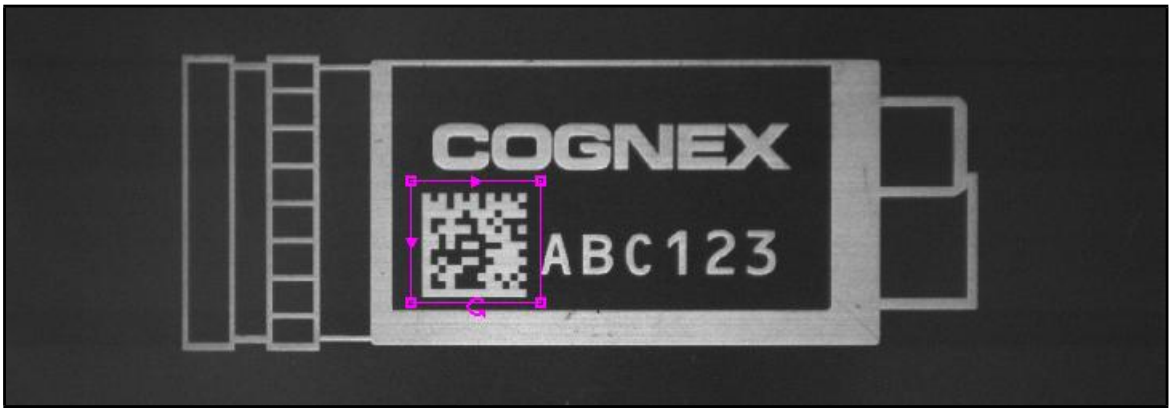
Application Step.
3. Expand the **Identification Tools** and select **Read 2D Code**. Click the **Add** 

button.





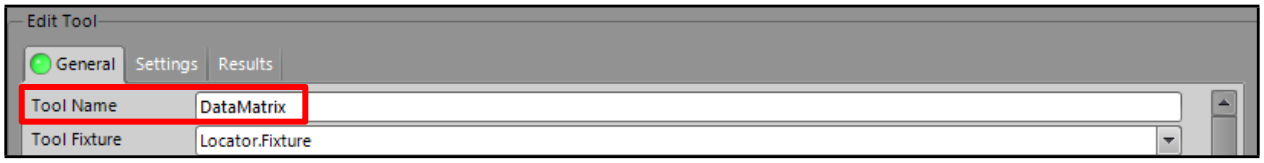
4. Place the tool region around the 2D Matrix feature to be inspected.



5. Click the **OK** OK button.

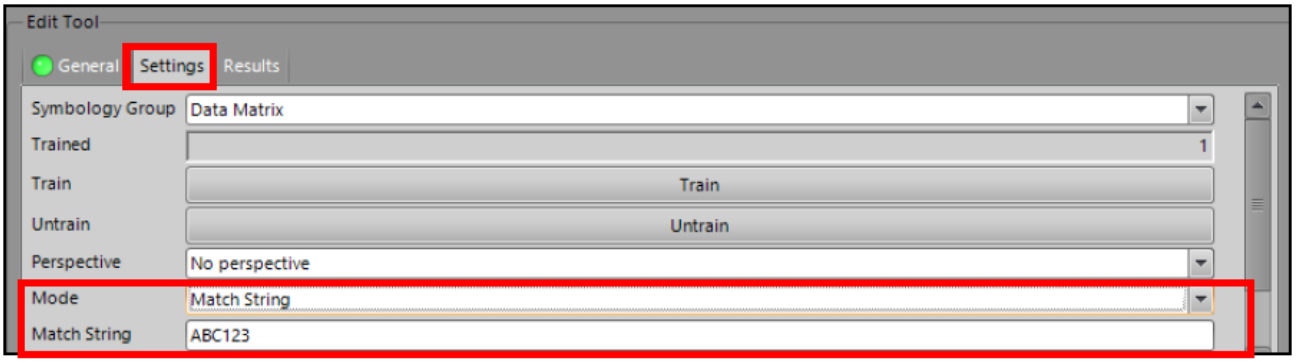
   NOTE: *Make sure the PASS image is showing.*

6. Change the **Tool Name** from IDCode_1 to DataMatrix.

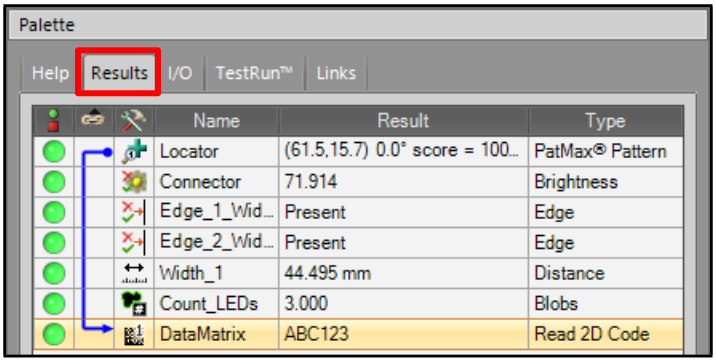


7. Click the **Settings** tab, change the **Mode** to **Match String** and confirm that **ABC123** is in the Match String field.

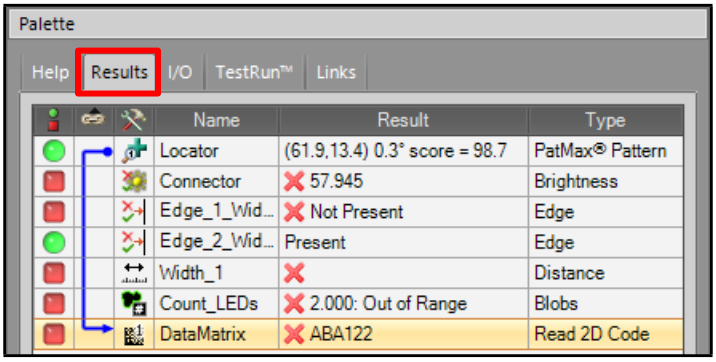
   NOTE: *The Match String will be set to the currently decoded string automatically.*



8. On the **Results** tab in the Palette confirm the tool is passing with a result of **ABC123**.

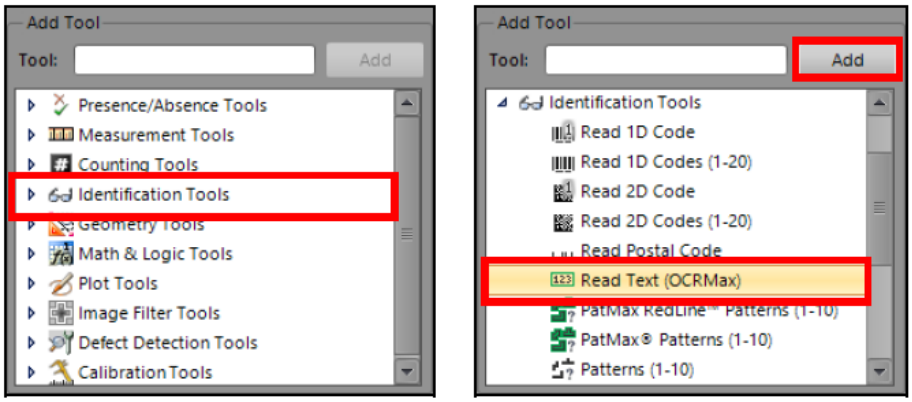


9. Flip the plate to the **FAIL** side, acquire a new image and confirm the tool is failing with a result of **ABA122**.



NOTE: *The tool fails because the decoded string does not match the Match String specified in Step 7.*

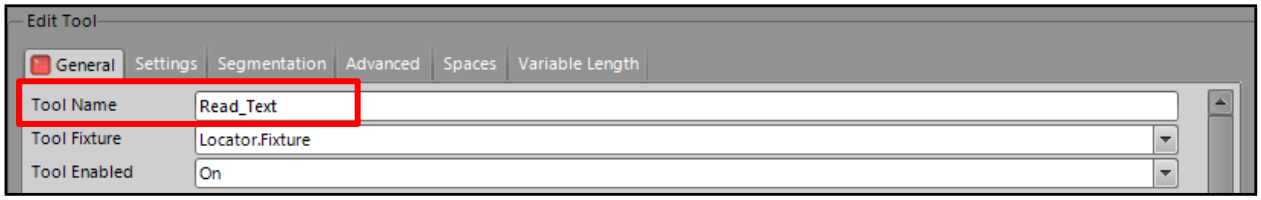
10. Expand the **Identification Tools** in the Inspection Tools and select **Read Text (OCRMax)**. Click the **Add** Add button.



11. Place the Rectangle tool around the alphanumeric string to be inspected and click the **OK** OK button.

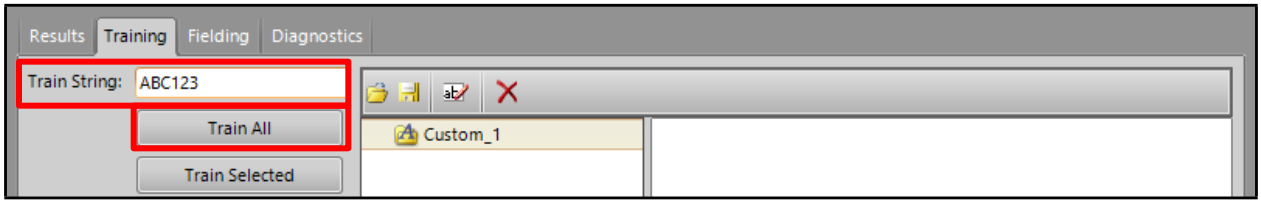


12. Change the Tool Name from Text_1 to **Read_Text.**

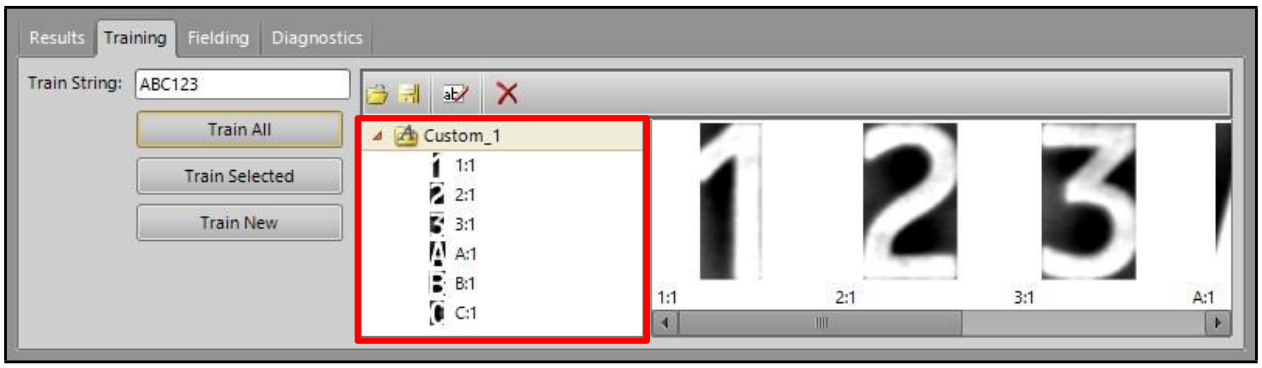


NOTE: *The tool is not passing – this is because we have not entered the string to be read.*

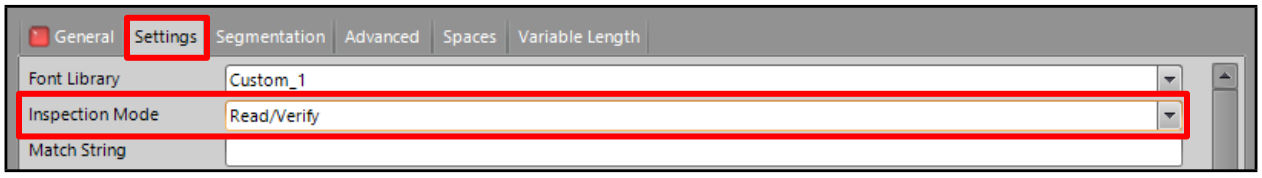
13. On the **Training** tab enter the string *ABC123* in the Train String field and click the **Train All** button.



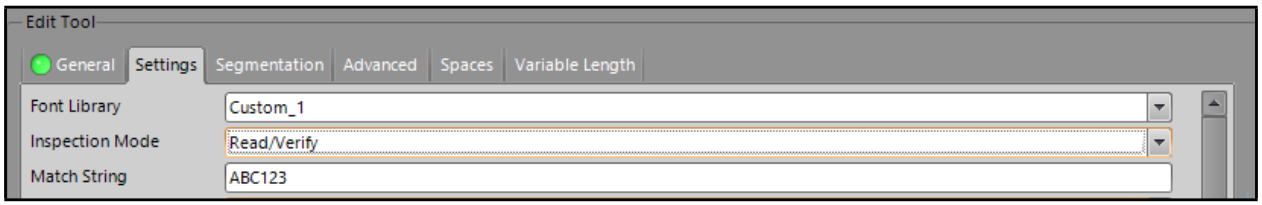
The characters are added to the custom library.



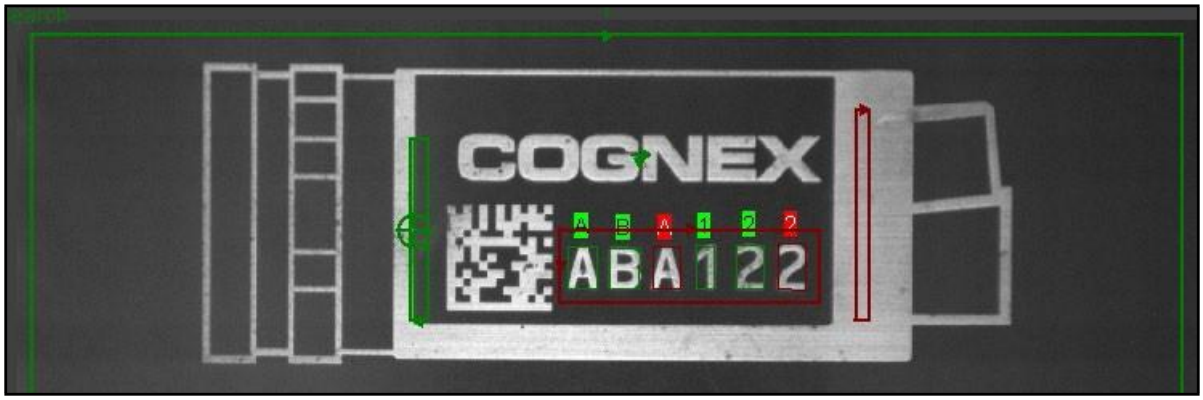
14. On the **Settings** tab change the **Inspection Mode** to **Read/Verify**. The tool now fails.



15. Enter the string *ABC123* in the **Match String** field and press the **<Enter>** key. The tool now passes.



16. Turn the plate over to the FAIL side and acquire a new image. Confirm the Read_Text tool is failing with a result of **ABA122**.



17. Save the job.

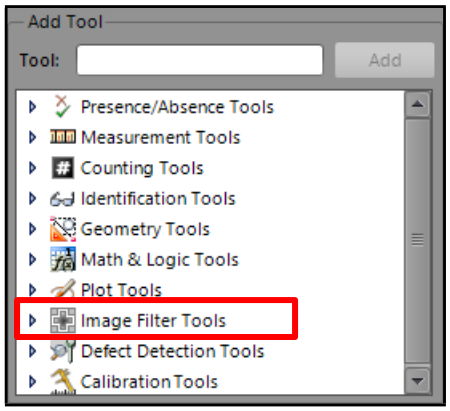
Enhance Text with Image Filter

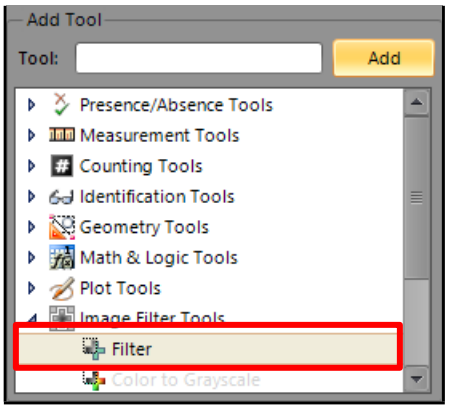
1. Continue with the **My2D** job, confirm that your demo plate is on the **PASS** side and that you are *not* in Live mode.

   Continue the **Inspect Part** 

Application Step.
2. Save the job as **MyFilter** in the folder created in Lab #2.
3. Expand the **Image Filter Tools** and select **Filter**. Click the **Add** 

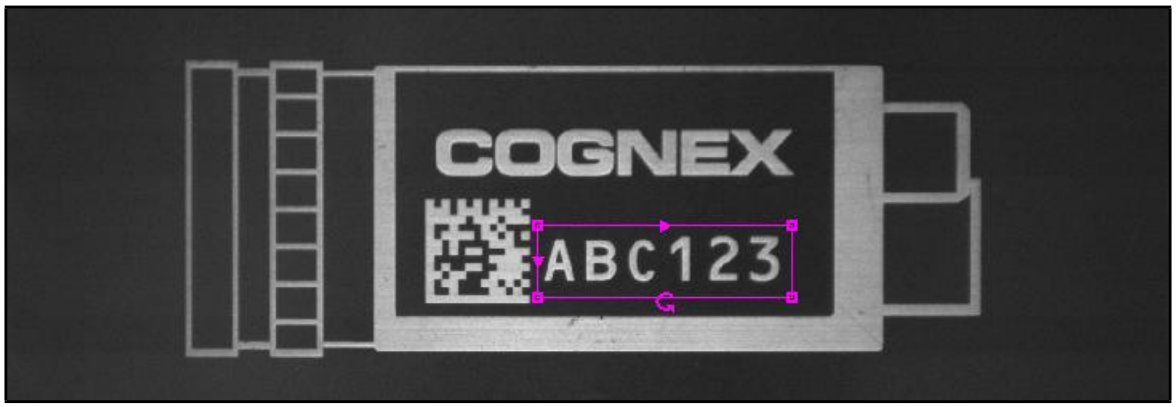
button.





4. Place the tool region around the Text area of the demo plate and click the **OK** 

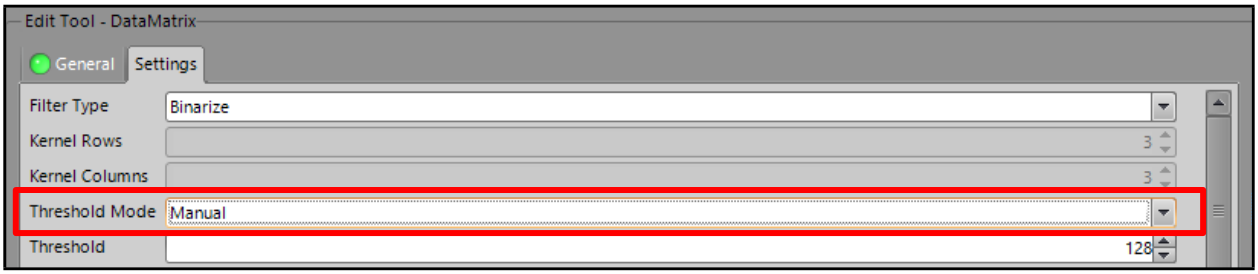
button.



5. Change the **Tool Name** from Filter_1 to *Binarize*.

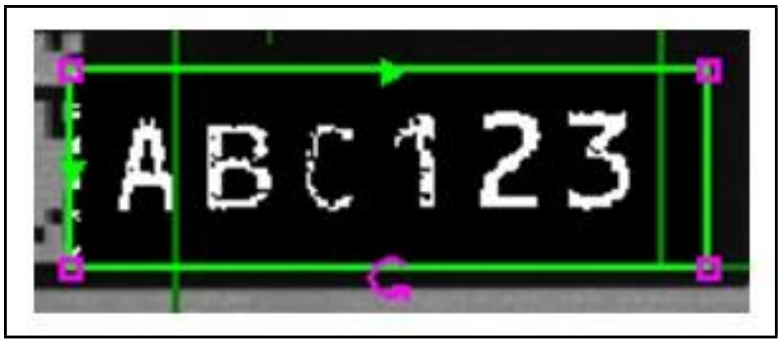


6. Click the **Settings** tab and change the **Threshold Mode** to **Manual**.



NOTE: *Adjust the Threshold until you get a good image.*

Original Image



Adjusted Image



7. Use the adjusted image for the Read_Text tool you added before by clicking the Read_Text tool and selecting the adjusted image for the **Tool Image** parameter under **General Tap**.
8. Save the job.

Lab 5.2 – Exercises

1. Use In-Sight 5400 emulation. Start a new job.
2. Browse to (or Open) the images file in the **OCRMax_Plain String** folder.
3. Assume that the characters **JV0** are present on every part.
4. Develop the job to read/verify the string in every image.
5. Save your job as **MyPlainString_YourName.job**. **Take a screenshot of the image view and the Palette. Submit the job and the screenshot on Blackboard**.
6. Hints: Under **Spaces Tap** – **Find Spaces**, select **Insert Single Space** to have the tool account for spacing. Enter the string in the **Train String box**, don't forget the space between **5** and **E**.

